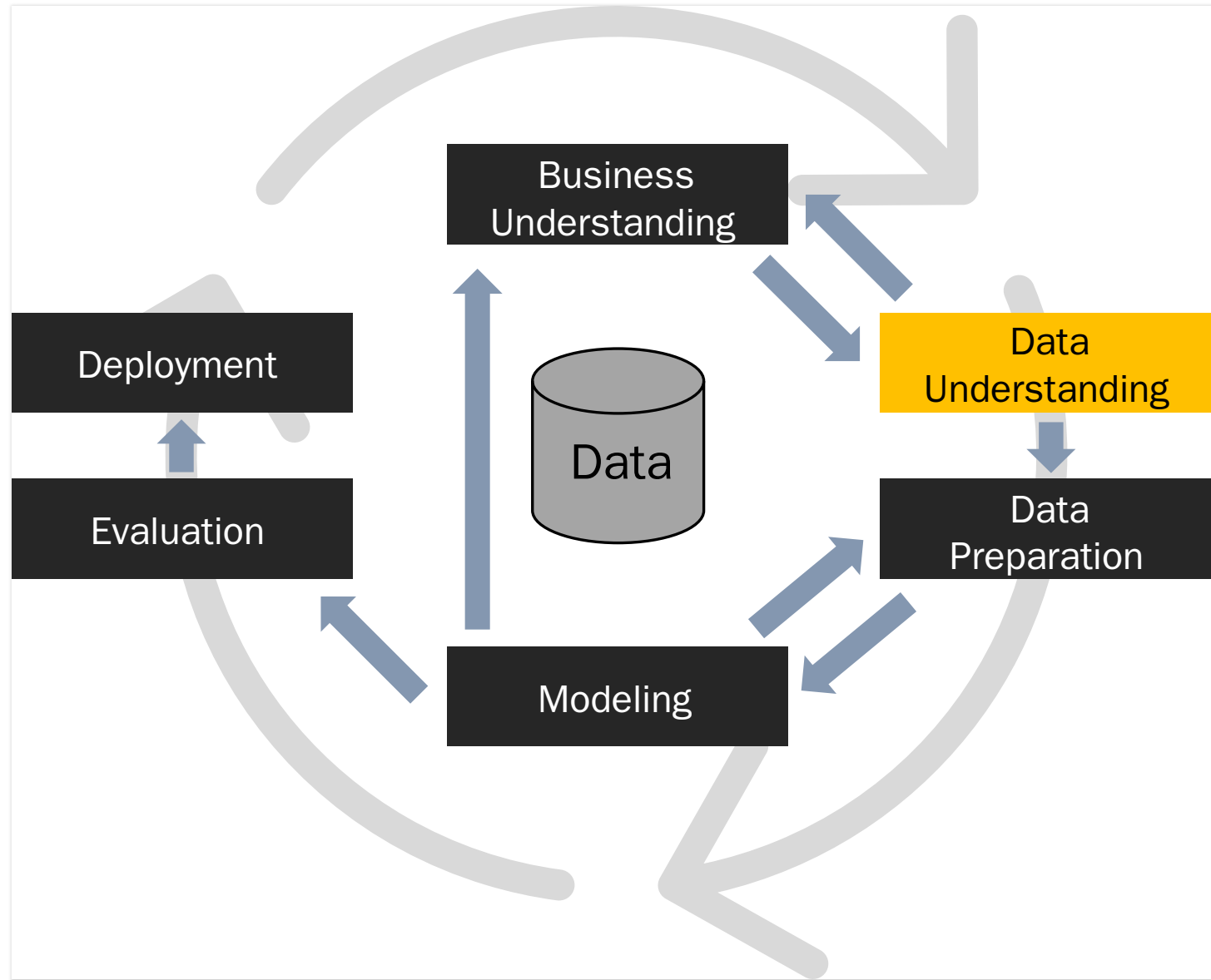


Machine Learning and Data Mining (Module 2)

Data Understanding

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Data understanding

The **data understanding** phase of CRISP-DM involves taking a closer look at the data available for mining.

- This step is critical in preventing problems during data preparation, which is typically the longest part of a project
- The data understanding phase involves four steps, including:
 1. *collection* of initial data
 2. *description* of data
 3. *exploration* of data, and
 4. *verification* of data quality

Data collection (or acquisition)

Data collection is the process of *gathering information on targeted variables* in an established system

- Capture evidence that allows data analysis to formulate credible answers to the questions that have been posed
- The analyst should make sure to report problems and solutions to aid with future replications of the project.
- Data may have to be collected from several different sources, and some of these sources may have a lag time.

The analyst then proceeds to

- increase familiarity with the data,
- identify data quality problems,
- discover initial insights into the data,
- detect interesting subsets to form hypotheses about hidden information



Problem: is data the new oil?

The more data we have, the more analysis we can do (however, more data != smarter data)

- There are several disciplines focusing on data (e.g., Data Science, Data Mining, Big Data, Business Intelligence)
- In Europe (but now in many areas of the world), there can be problems related to privacy
 - When is it “right” to protect privacy?
 - When does it become a limit?

Acquiring data is a time-consuming, investment and knowledge-intensive process

- How much data is *enough*?

Rule of thumb: *one in ten/twenty* ([Chowdhury and Turin 2020](#))

There is no set rule as to the number of variables to include in a prediction model as it often depends on several factors. The ‘one in ten rule’, a rule that stipulates how many variables/parameters can be estimated from a data set, is quite popular in traditional clinical prediction modeling strategies (e.g., logistic regression and survival models). According to this rule, one variable can be considered in a model for every 10 events.



Problem: how can we collect data?

Acquisition of a new dataset

Data collection is usually a costly process.

Investment of time and money for:

- Programming or learning to use an acquisition tool
- Handling of *large amounts of data*
- Testing to find any bugs that could compromise the success of the acquisition
 - Unfortunately, we often notice them at the end of the process
- *Acquire new hardware* for data collection and storage

It is necessary to carefully consider whether it is appropriate to acquire a new dataset.

- Considerations not only in engineering but also in management and economic aspects
- Future needs must be foreseen in advance

Common Problems in Data Collection

Sometimes companies have a lot of “old” data in their databases or information systems:

- They don't know what to do with it
- Data re-collection on existing data (since data must be clean or something similar)

In many business processes, it is unclear:

- Which data is possible to collect (also due to privacy issues)
- Which data is (really) useful for the business

Public Datasets

We can get data mainly in two ways.

By using *publicly available data* (datasets or databases)

- Someone collected them for us!
- They can be free or for a fee
- The quality of the data made available must be checked

By *acquiring a new set of data*, but why?

- It is not certain that public data well represents the problem we want to solve
- We want to acquire specific data and thus generate specific expertise for the company (know-how)
- We are forced to acquire data that, due to their sensitive nature, would not otherwise be available (privacy issues)
- The company we work for already has a data collection process that we can use

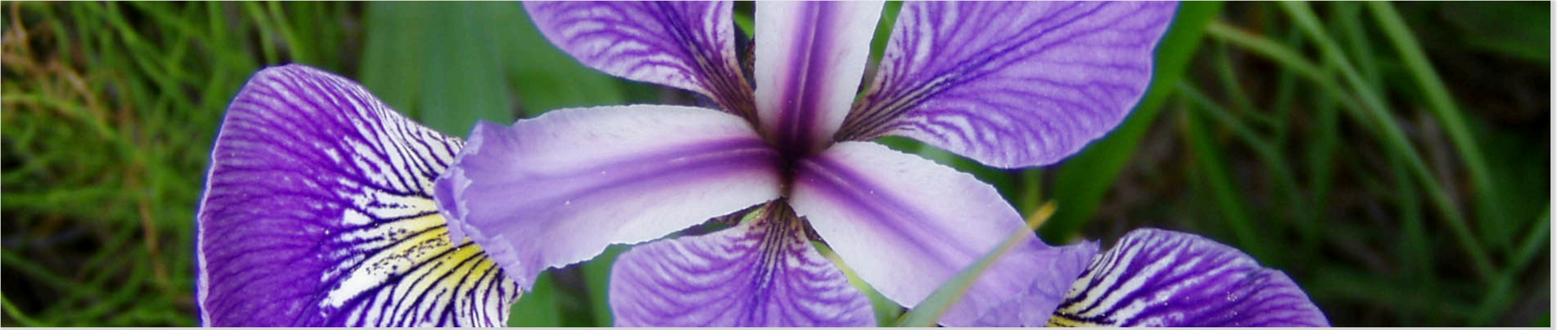
Public Datasets

Many universities publicly release their datasets:

- There are no requirements related to profit or non-disclosure agreement (NDA)
- It is the basis of the scientific method, in particular for the reproducibility of the results obtained
- I release my data so that others can conduct my experiments and verify my results
- Examples: <https://www.image-net.org/>

Some platforms make datasets available for competitions, such as [Kaggle](#) and [others](#)

Example of a public dataset: the Iris dataset



The [Iris dataset](#) is public data that was used in R.A. Fisher's classic 1936 paper ([Fisher 1936](#))

- It can also be found on the [UCI Machine Learning Repository](#).
- It includes 3 iris species (Setosa, Virginica, and Versicolor) with 50 samples each
- It characterizes flowers with some properties of each flower
 1. `SepalLengthCm`
 2. `SepalWidthCm`
 3. `PetalLengthCm`
 4. `PetalWidthCm`
 5. `Species`



Problem: now that we have the data, what do we do?

Describe the Data

The key question to ask is: does the data acquired satisfy the relevant requirements?

- This step also provides a basic understanding of the data on which subsequent steps will be built.
- For instance, if age is important and the data does not reflect the entire age range, it may be wise to collect a different dataset

The data analyst examines the “surface” properties of the acquired data, examining issues such as:

- the *format* of the data,
- the *quantity* of the data,
- the *number of records and fields* in each table,
- the *identities* of the fields,
- and any other *surface features of the data*.

Iris

Plain Iris dataset

Id	SepalLengthCm	SepalWidthCm	PetalLengthCm	PetalWidthCm	Species
1	5.1	3.5	1.4	0.2	Iris-setosa
2	4.9	3	1.4	0.2	Iris-setosa
3	4.7	3.2	1.3	0.2	Iris-setosa
4	4.6	3.1	1.5	0.2	Iris-setosa
5	5	3.6	1.4	0.2	Iris-setosa
...	...	...	...	...	...

Iris

Example of profiling the schema of the data in Iris

```
RangeIndex: 150 entries, 0 to 149
Data columns (total 5 columns):
#   Column              Non-Null Count  Dtype
---  -
0   sepal length (cm)    150 non-null    float64
1   sepal width (cm)     150 non-null    float64
2   petal length (cm)    150 non-null    float64
3   petal width (cm)     150 non-null    float64
4   species              150 non-null    object
dtypes: float64(4), object(1)
memory usage: 6.0+ KB
```

Iris

Example of profiling the distribution of the data in Iris

	Id	SepalLengthCm	SepalWidthCm	PetalLengthCm	PetalWidthCm
count	150	150	150	150	150
mean	75.5	5.84333	3.054	3.75867	1.19867
std	43.4454	0.828066	0.433594	1.76442	0.763161
min	1	4.3	2	1	0.1
25%	38.25	5.1	2.8	1.6	0.3
50%	75.5	5.8	3	4.35	1.3
75%	112.75	6.4	3.3	5.1	1.8
max	150	7.9	4.4	6.9	2.5

Iris

In descriptive statistics, a **box plot** shows graphically the locality, spread, and skewness of numerical data

A boxplot is a standardized way of displaying the dataset based on the five-number summary:

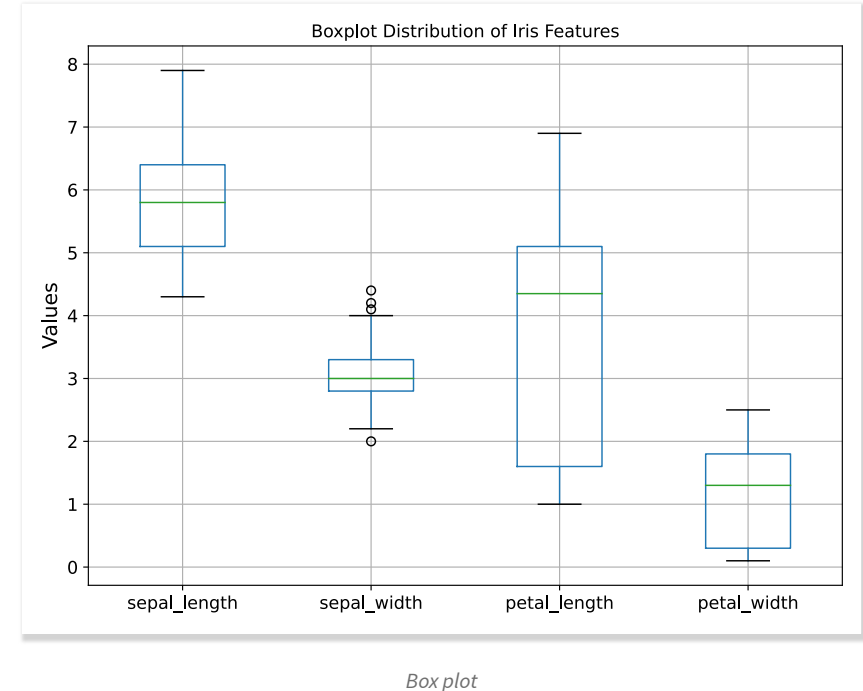
1. *Minimum* (Q_0 or 0th percentile): the lowest data point
2. *First quartile* (Q_1 or 25th percentile)
3. *Median* (Q_2 or 50th percentile): the middle value
4. *Third quartile* (Q_3 or 75th percentile)
5. *Maximum* (Q_4 or 100th percentile): the highest data point

Interquartile range: $IQR = Q_3 - Q_1$

Graphical elements

- The *box* is drawn from Q_1 to Q_3
- *Whiskers* are based on the $1.5 \cdot IQR$ value
 - A whisker is drawn up to the largest/lowest observed data point from the dataset that falls within this distance
 - The whisker lengths can look unequal
- All other points outside the whiskers are plotted as *outliers*

Iris data



Explore the Data

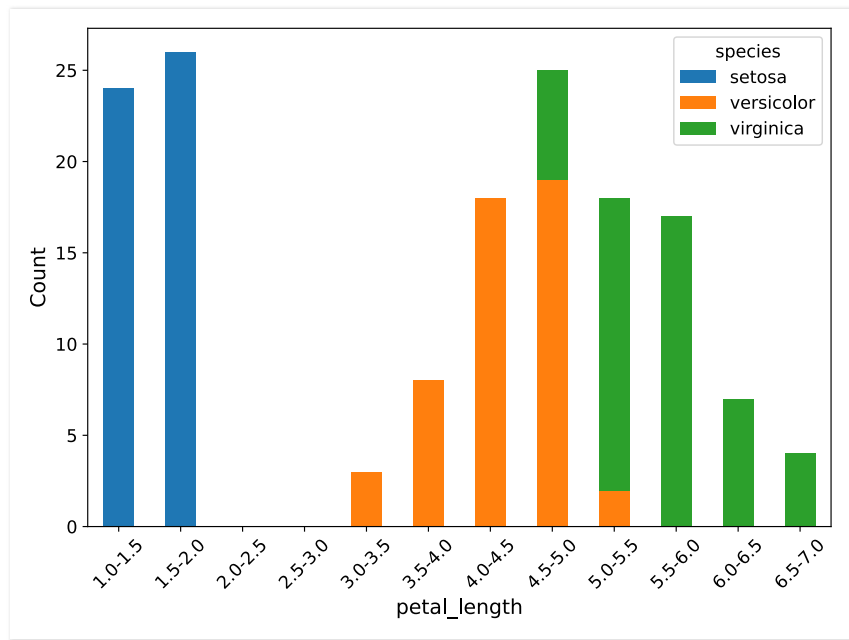
This task tackles the data mining questions, which can be addressed using querying, visualization, and reporting.

- Create a data exploration report that outlines the first findings, or an initial hypothesis
- For instance, query the data to discover which features are correlated with the target of the analysis

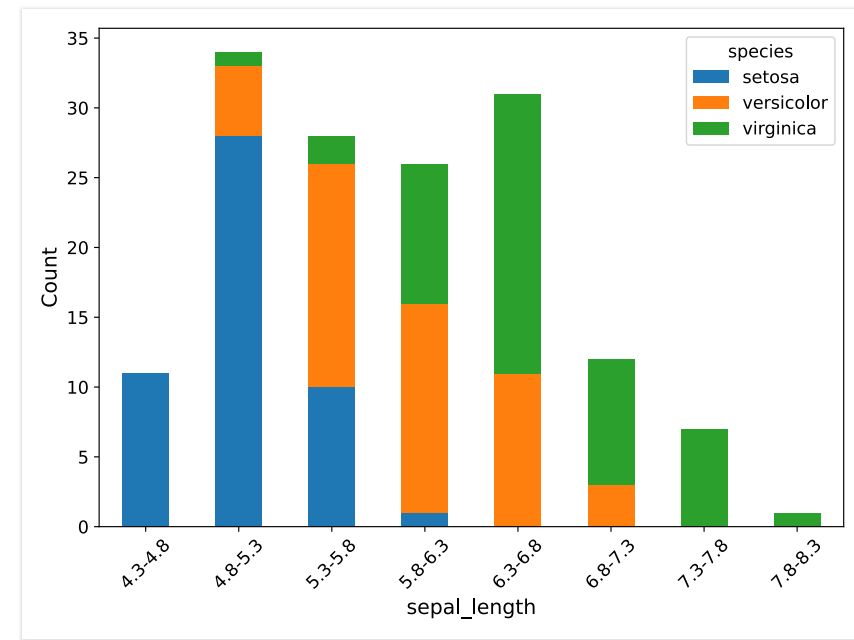
Iris

Value distribution vs `species`

`petal_length` vs `species`

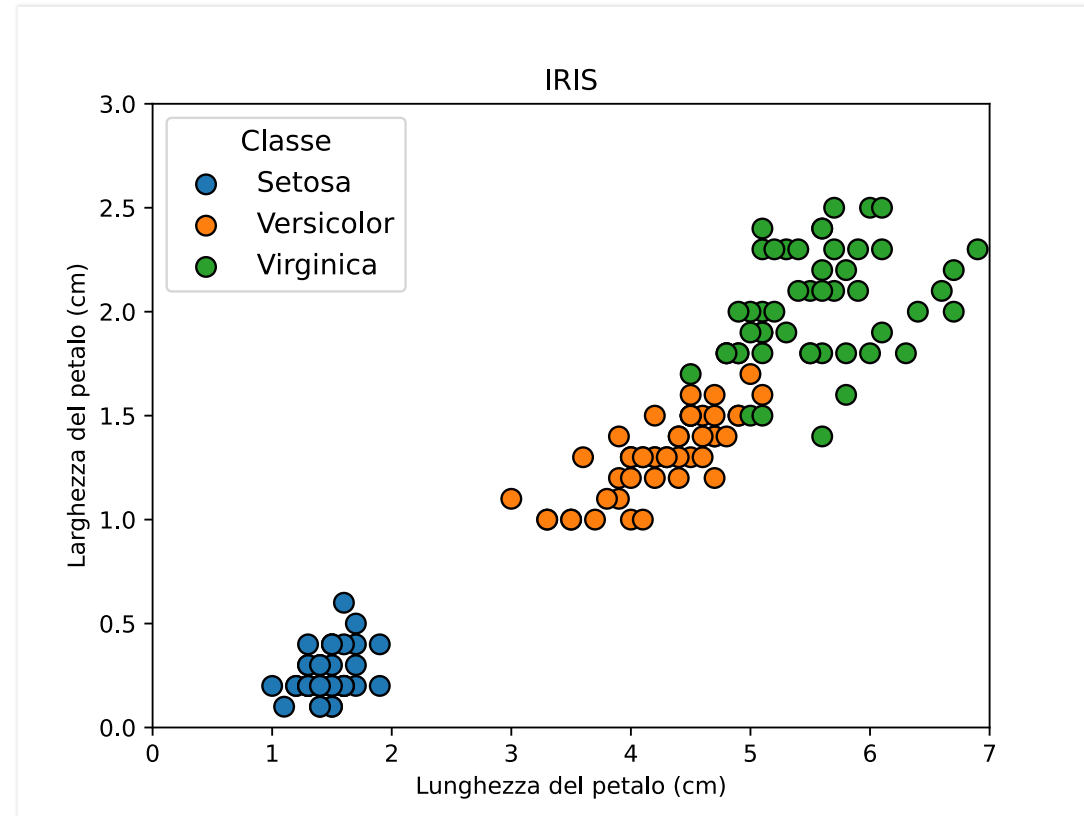


`sepal_length` vs `species`



Iris

2D visualization of the Iris dataset



Verify Data Quality

At this point, the analyst examines the quality of the data, addressing questions such as:

Some common data quality issues to check include:

- *missing values* or attributes;
 - E.g., unknown date of death (Is it missing? Is the person alive?)
- whether *all possible values are represented*;
 - E.g., all age groups are contained in the dataset
- the *plausibility of values*, review any attributes that may give answers that conflict with common sense;
 - E.g., teenagers with high income
- whether attributes with different values have *similar meanings*;
 - E.g., `low fat` and `diet`
- the *spelling* of values.
 - E.g., `law fat` or `low fat`?

Nothing to worry about in Iris

Dimensions of Data Quality (Sidi et al. 2012)

Types of Data	Definition
Consistency	The extent to which data is presented in the same format and compatible with previous data [15]. Refer to the violation of semantic rules defined over the set of data [2].
Accuracy	Data are accurate when data values stored in the database correspond to real-world values [2, 19]. The extent which data is correct, reliable and certified [15]. Accuracy is a measure of the proximity of a data value, v, to some other value, v', that is considered correct [2, 17]. A measure of the correction of the data (which requires an authoritative source of reference to be identified and accessible [14]).
Completeness	The ability of an information system to represent every meaningful state of the represented real world system [2, 11]. The extent to which data are of sufficient breadth, depth and scope for the task at hand [15]. The degree to which values are present in a data collection [2, 17]. Percentage of the real-world information entered in the sources and/or the data warehouse [2, 18]. Information having all having all required parts of an entity's information present [2, 16]. Ratio between the number of non-null values in a source and the size of the universal relation [2, 20]. All values that are supposed to be collected as per a collection theory [2, 21].

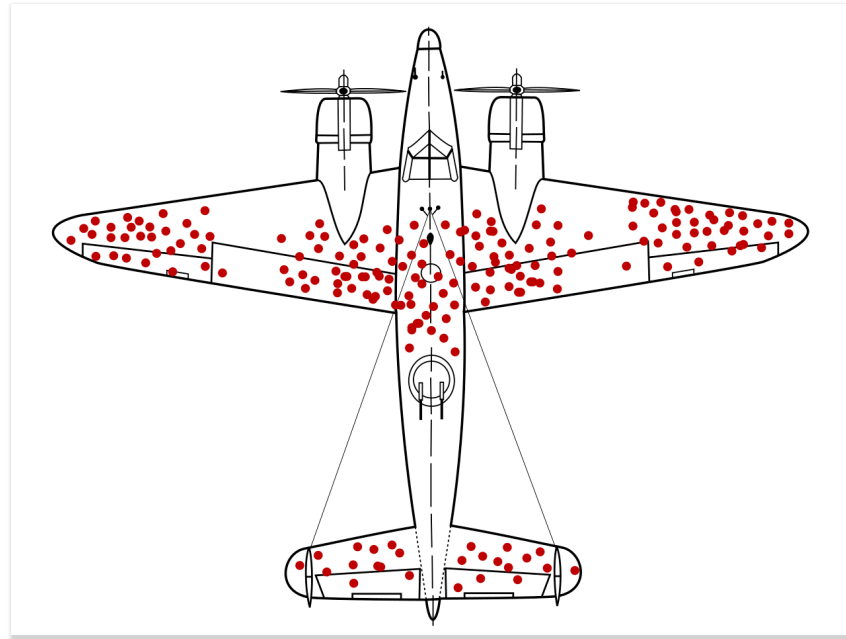
Dimension	Definition
Timeliness	The extent to which age of the data is appropriated for the task at hand [15]. Timeliness refers only to the delay between a change of a real world state and the resulting modification of the information system state [2, 11]. Timeliness has two components: age and volatility. Age or currency is a measure of how old the information is, based on how long age it was recorded. Volatility is a measure of information instability the frequency of change of the value for an entity attribute [2, 16].
Currency	Currency is the degree to which a datum is up-to-date. A datum value is up-to-date if it is correct is spite of possible discrepancies caused by time-related changes to the correct value [2, 17]. Currency describes when the information was entered in the sources and/or the data warehouse. Volatility describes the time period for which information is valid in the real world [2, 18].

Metrics of Data Quality (Batini et al. 2009)

Table IX. Dimensions and Metrics		
Dimensions	Name	Metrics Definition
Accuracy	Acc1	Syntactic accuracy: it is measured as the distance between the value stored in the database and the correct one Syntactic Accuracy = Number of correct values/number of total values
	Acc2	Number of delivered accurate tuples
	Acc3	User Survey - Questionnaire
Completeness	Compl1	Completeness = Number of not null values/total number of values
	Compl2	Completeness = Number of tuples delivered/Expected number
	Compl3	Completeness of Web data = $(T_{max} - T_{current}) * (Completeness_{Max} - Completeness_{Current}) / 2$
	Compl4	User Survey - Questionnaire
Consistency	Cons1	Consistency = Number of consistent values/number of total values
	Cons2	Number of tuples violating constraints, number of coding differences
	Cons3	Number of pages with style guide deviation
	Cons4	User Survey - Questionnaire
Timeliness	Time1	Timeliness = $(\max(0; 1 - \text{Currency}/\text{Volatility}))^s$
	Time2	Percentage of process executions able to be performed within the required time frame
	Time3	User Survey - Questionnaire
Currency	Curr1	Currency = Time in which data are stored in the system - time in which data are updated in the real world
	Curr2	Time of last update
	Curr3	Currency = Request time- last update
	Curr4	Currency = Age + (Delivery time- Input time)
	Curr5	User Survey - Questionnaire
Volatility	Vol1	Time length for which data remain valid
Uniqueness	Uni1	Number of duplicates
Appropriate amount of data	Appr1	Appropriate Amount of data = $\text{Min}((\text{Number of data units provided}/\text{Number of data units needed}); (\text{Number of data units needed}/\text{Number of data units provided}))$



Problem: what about biases?



Red dots stand for places where surviving planes were shot. How would you reinforce the planes?

Biases

Bias

- Inclination or prejudice for or against one person or group, especially in a way considered to be unfair.
 - “There was evidence of bias against foreign applicants.”
- A systematic distortion of a statistical result due to a factor not allowed for in its derivation.
 - “The search results are biased by the specific queries used.”

During data collection and analysis, several **biases** can occur

- *Selection*: sample used for data collection is not representative of the population being studied
- *Sampling*: certain segments of the population are more likely to be included or excluded from the sample
- *Response*: participants in a survey or study provide inaccurate or misleading responses
- *Confirmation*: refers to the tendency to favor information that confirms pre-existing beliefs or hypotheses while ignoring contradictory evidence
- *Cultural*: data collection methods, survey questions, or study designs are culturally insensitive or fail to account for cultural differences
- *Time-Interval*: the timing of data collection influences the results
- ... and many others

Confirmation bias

“If you torture the data long enough, it will confess to anything”

Ronald H. Coase

(2023) Dati “torturati” per dare un assist al governo

Survivorship bias

The “survivors” get studied, while the failures are excluded, leading to potentially flawed conclusions.

Start-up Success Stories

People often hear stories of wildly successful companies like Apple, Amazon, or Tesla and assume that hard work and a good idea are enough to succeed. This overlooks the countless failed start-ups that had hard-working teams and great ideas but didn’t survive due to market conditions, competition, or other factors.

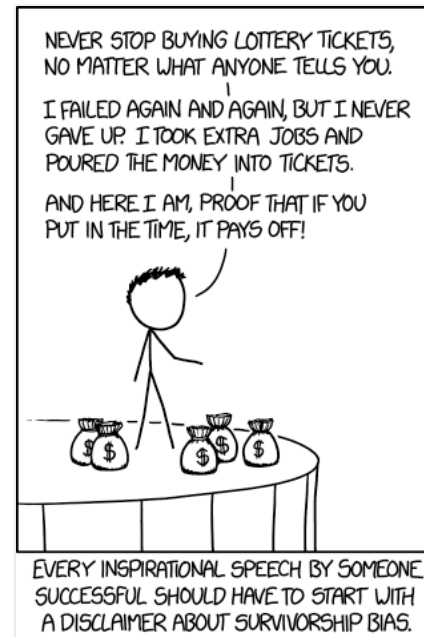
Fitness and Weight Loss Programs

Testimonials for fitness programs often highlight people who achieved dramatic results. These “success stories” ignore the many individuals who followed the same program but didn’t achieve noticeable results, either due to differences in genetics, lifestyle, or other factors.

Investment Portfolios

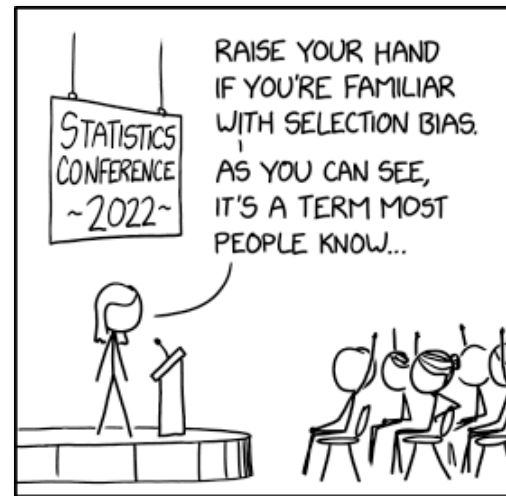
Financial advice often highlights top-performing stocks or mutual funds as examples of great investments. These examples focus on the “survivors” in the market, ignoring the many investments that failed or underperformed, which can lead to overestimating the likelihood of similar success in the future.

Survivorship bias



<https://xkcd.com/1827/>

Selection bias



<https://xkcd.com/2618/>

... and more!

Confirmation Bias

A person who believes in a specific political ideology might seek out news articles or social media posts that reinforce their beliefs, while ignoring or dismissing information that contradicts their views. This selective exposure to information strengthens their pre-existing opinions.

Anchoring Bias

When shopping for a new car, a person may see a car priced at \$30,000, and then a second car priced at \$25,000. Even if the second car is not objectively better or a great deal, the first price “anchors” their perception of the value of the second car, leading them to think it’s a better deal simply because it’s cheaper than the first option.

Availability Bias

After watching several news reports about airplane crashes, a person might overestimate the risk of flying. They might avoid flying, despite it being statistically safer than driving, because the images and stories of crashes are more readily available in their memory.

Biases and fairness

Fairness refers to the various attempts to correct algorithmic bias in automated decision processes based on ML models.

- Decisions made by such models after a learning process may be considered unfair if they were based on variables considered sensitive
- E.g., gender, ethnicity, sexual orientation, or disability

[\(2018\) Amazon scraps secret AI recruiting tool that showed bias against women](#)

How could we approach this problem?

Biases and fairness

(2024) [‘We definitely messed up’: why did Google AI tool make offensive historical images?](#)

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